

Erratum: “Unified Microscopic Foundations of the Quantum Boltzmann Equation”

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We have identified sign- and factor-level errors in the intermediate steps of the Green-function derivation of Sec. 3, together with a factor-of-two error in the Markov step of Appendix A that propagated into the coefficient of the final rate expressions. All equation, figure, and page numbers below refer to the published article. None of the corrections affects the structure of the Uehling–Uhlenbeck collision integral, the quantum-statistical factors, the equilibrium solutions, the H -theorem, or any conclusion of the article.

1. Green-function derivation (Sec. 3)

Equation (19) is a projection rule for contour *convolutions*, but X and Y of Eq. (15) are not convolutions: the interaction is instantaneous, so the internal times are pinned to the external ones ($t_2 = t_1$, $t_{2'} = t_{1'}$) and X , Y are pointwise products of contour functions. Equations (20)–(23) and (26) then follow from the branch assignment of each factor alone, with no retarded or advanced propagators.

1. Equations (20) and (21) should read

$$X^{\lessgtr}(1, 1') = G^{\lessgtr}(1, 1') G^{\lessgtr}(2, 2') G^{\lessgtr}(2', 2), \quad Y^{\lessgtr}(1, 1') = G^{\lessgtr}(1, 2') G^{\lessgtr}(2, 1') G^{\lessgtr}(2', 2),$$

giving the lesser and greater components together, so that the following sentence, “Similarly, we can write for $X^>(1, 1')$ and $Y^>(1, 1')$ ”, should be deleted.

2. In Eq. (22) the Green-function product in the integrand should read $G^<(1, 1') G^<(2, 2') G^>(2', 2)$, and in Eq. (23) it should read $G^<(1, 2') G^<(2, 1') G^>(2', 2)$. The justification preceding each equation, which selects “one lesser, two greater” terms in a Born and quasiparticle limit, should instead note that the projection is direct: the lesser (greater) self-energy carries two lesser (greater) propagators and one greater (lesser).
3. In Eq. (26) the square bracket should read $G^<(1, 1') G^<(2, 2') G^>(2', 2) + \eta G^<(1, 2') G^<(2, 1') G^>(2', 2)$, with the statistics factor η in place of the published $\pm$.
4. In Eqs. (35) and (37) the lesser/greater superscripts on the collision terms are interchanged and a factor i is missing. The right-hand sides should read $i[\Sigma^<G^> - \Sigma^>G^<]$, with Σ_{coll} in Eq. (37). Correspondingly, the bracket of Eq. (40) should read $\Sigma^<(\mathbf{p}, \omega) G^>(\mathbf{p}, \omega) - \Sigma^>(\mathbf{p}, \omega) G^<(\mathbf{p}, \omega)$: the gain term is $\Sigma^<G^>$, the loss term $\Sigma^>G^<$.
5. Equations (44) and (45) omit the factors $\pm i$ in their prefactors and interchange the in- and out-scattering occupation structures. They should read

$$\Sigma_c^{\lessgtr}(\mathbf{p}, \omega_{\mathbf{p}}) = (\pm 2\pi i) \sum_{\mathbf{p}_1, \mathbf{p}'_1, \mathbf{p}'_2} U(\mathbf{p} - \mathbf{p}_1) U(\mathbf{p}'_2 - \mathbf{p}'_1) \delta(\Delta\omega) F^{\lessgtr},$$

with $\Delta\omega = \omega_{\mathbf{p}} + \omega_{\mathbf{p}_1} - \omega_{\mathbf{p}'_1} - \omega_{\mathbf{p}'_2}$, $F^< = (1 + \eta n_{\mathbf{p}_1}) n_{\mathbf{p}'_1} n_{\mathbf{p}'_2}$ and $F^> = n_{\mathbf{p}_1} (1 + \eta n_{\mathbf{p}'_1}) (1 + \eta n_{\mathbf{p}'_2})$. The prefactors $\pm 2\pi i$ mirror the spectral factors of Eq. (38), and the chain then reproduces Eq. (46) exactly. These are the *direct-channel* self-energies, and the sentence preceding Eq. (44) should say so.

2. Overall coefficient: Appendix A and Eqs. (5)–(7), (47)

In the Markov step of Appendix A the long-time limit of the time integral is correctly given in the text as $\pi\delta(\Delta) + i\mathcal{P}(1/\Delta)$, but the coefficient carried forward corresponds to $2\pi\delta(\Delta)$. The four operator families of the second-order bracket form two Hermitian-conjugate pairs, between which the principal-value contributions cancel, leaving each family the weight $\pi\delta$.

Correcting this halves the coefficient of the second-order term in every equation of the chain that follows: Eqs. (A.2), (A.3), and (A.5)–(A.7) of the appendix, and Eqs. (5)–(7) and (47) of the main text, so that the final rate expressions (7) and (47) carry $-4\pi/\hbar^2$ rather than the published $-8\pi/\hbar^2$. Nothing else changes, and (7) and (47) continue to agree, so the two derivations remain consistent.

3. The Wick-pairing count

The article states that a six-field correlator has fifteen pairings, three of which are Hartree–Fock, “leaving twelve nontrivial contractions”. This conflates pair-partitions of the correlator with the real-time channels

they generate. Of the fifteen pair-partitions, nine vanish for a number-conserving theory, three contain an equal-point contraction $\langle \psi^\dagger(a)\psi(a) \rangle = n(a)$ and give the mean-field (Hartree–Fock) contributions, and one, $G(1,2)G(2,2')G(2',1')$, is a reducible iteration of the Fock mean field, absorbed into the mean-field renormalization of the dispersion. The remaining two are the direct (X) and exchange (Y) correlation topologies, and it is these that generate the twelve real-time gain–loss channels on contour ordering and $\leq$ projection. The count of *twelve real-time channels*, on which the article’s central correspondence rests, is unchanged.

The sentences stating this count should be amended accordingly in the Abstract and on pp. 2640002-2, -3, -5, -6 (twice), -8, -9 (twice), -11 (twice), and -16, where the several phrasings of “twelve Wick contractions” should each read “twelve real-time channels”. Each topology has one lesser and one greater component, not “three lesser and three greater” (p. 2640002-9), so the count $3 + 3 + 3 + 3 = 12$ (p. 2640002-11) should read that $X^{\leq}$ and $Y^{\leq}$ reproduce the twelve occupation monomials of Eq. (5), six per branch.

Relatedly, the six-field correlator displayed on p. 2640002-9 separates the two fields of each internal vertex, leaving their equal-time contour ordering unspecified. It should read $\langle T_C \psi(1) \psi^\dagger(2) \psi(2) \psi^\dagger(2') \psi(2') \psi^\dagger(1') \rangle$, each internal vertex contributing a density pair $\psi^\dagger \psi$ at its own point.

4. Further corrections

- **p. 2640002-7.** The parenthetical after Eq. (7) should state that Eqs. (6) and (7) are algebraically identical, since the quartic η^2 terms cancel pairwise on expansion, rather than that they differ by an antisymmetry vanishing on summation.
- **Fig. 1** (p. 2640002-9). In the exchange (Y) panel the propagator joining vertices $2'$ and 2 is labeled $G(2,2')$. It should be $G(2',2)$, as in the expression printed beneath the panel and in Eq. (15).
- **p. 2640002-13.** “Multiplying Eq. (34) by $(-i)$ ” should read “Multiplying the Wigner-transformed kinetic equation (35), specialized to a spatially homogeneous system, by $(-i)$ ”. Equation (34) is the two-time equation, not its Wigner representation.
- **p. 2640002-14.** The sentence preceding Eq. (43) should state that momentum and energy conservation together imply the equality of the relative-momentum *magnitudes*, $|\mathbf{p} - \mathbf{p}_1| = |\mathbf{p}'_1 - \mathbf{p}'_2|$, with $\mathbf{q} \equiv \mathbf{p} - \mathbf{p}_1$ the relative momentum, and that for an isotropic kernel, $U = U(|\mathbf{q}|)$, this renders the product (42) the perfect square $|U(\mathbf{q})|^2$ of Eq. (43). The printed vector identity $\mathbf{p} - \mathbf{p}_1 = \mathbf{p}'_2 - \mathbf{p}'_1$ does not follow from momentum conservation (the two sides coincide only when $\mathbf{p}'_1 = \mathbf{p}_1$).
- **p. 2640002-19.** “commutation relations (2.1)” should read “the commutation relation quoted after Eq. (4)”. The label (2.1) is not a numbered equation of the article.
- **Reference [12].** The entry should read: M. Bonitz, *Quantum Kinetic Theory*, 2nd edn. (Springer, Cham, 2016). The printed volume designation “Vol. 412” does not apply, and the edition was omitted.

The authors regret these errors.